

Price Volatility

INTRODUCTION

In this chapter, we discuss price volatility and factor models and how they can be used to improve trading performance. We present various techniques that are used in the industry to forecast volatility as well as appropriate methods to calibrate these models.

Volatility is the uncertainty surrounding potential price movement, calculated as the standard deviation of price returns. It is a measure of the potential variation in price trend and not a measure of the actual price trend. For example, two stocks could have the exact same volatility but much different trends. If stock A has volatility of 10% and price trend of 20%, its one standard deviation return will be between 10 and 30%. If stock B also has volatility of 10% but price trend of 5%, its one standard deviation return will be between -5 and 15%. Stock with higher volatility will have larger swings than the stock with lower volatility, resulting in either higher or lower returns.

There are two volatility measures commonly used in the industry: realized and implied. Realized volatility is computed from historical prices and is often referred to as historical volatility. Realized volatility uses past history to predict the future. Implied volatility, on the other hand, is computed from the market's consensus of the fair value for a derivative instrument such as the S&P500 index option contract. Implied volatility is a “forward” looking or “future” expectation estimate.

Historical Volatility lets the data predict the future.

Implied Volatility lets the market predict the future.

We utilize volatility in many different ways. For example, traders use volatility to understand potential price movement over the trading day, as input into market impact models, to compute trading costs, and to select algorithms. Algorithms use volatility to determine when it is appropriate